

Scalable Admin, Doctor & Staff Onboarding

Agentic Implementation Plan for Antigravity CLI (agy)

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This plan is written to be executed **phase-by-phase by an Antigravity CLI agent**. Each phase lists exact files to touch, the changes required, acceptance criteria, and a verification command. **Do not start a phase until the previous phase's verification passes.**

Phase Overview

Phase	Focus	Key Deliverables
1	Data model foundations	is_verified flags, staff role, FacilityMembership
2	Self-serve registration	Facility + Doctor registration serializers & endpoints
3	Staff delegation	Scoped staff create/list/delete for Facility Admins
4	Verification Queue	1-click approve/reject + Platform Admin invite
5	Automated tests	test_onboarding, test_staff_delegation, test_verification
6	Frontend Registration Hub	Multi-tab Sign-In + Register forms on /admin
7	Staff + Queue tabs	StaffTab, VerificationQueueTab, nav role gating
8	Hardening	Throttling, public filters, audit log, cleanup

Critical Design Rules (Do Not Violate)

- Super Admin can **never** be created via public registration.
- Newly registered facilities and doctors start with **is_verified=False** and can log in immediately, but stay hidden from public search until verified.
- Facility Staff are created only by Facility Admins (or Super Admins) — never self-register.
- All registration endpoints must be rate-limited and must validate uniqueness (phone, BMDC, license numbers).
- Atomic transactions for every multi-model creation path.

Phase 1 — Data model foundations

Goal: Make the database ready for pending vs live entities and for Facility Staff.

Files to touch

• accounts/models.py • doctors/models.py • facilities/models.py • New migrations

Key Changes

User: Add is_verified BooleanField (default=False). Extend Role with STAFF. Update save() so staff role forces is_staff=False / is_superuser=False. Add is_staff_role property.

Doctor: Add bmdc_number (unique, nullable) and is_verified.

Location / FacilityMembership: Ensure Location has is_verified and location_type. Ensure FacilityMembership exists with ADMIN | STAFF roles and unique(user, location) constraint.

Verification

```
cd doctors_hub_backend
python manage.py makemigrations accounts doctors facilities
python manage.py migrate && python manage.py check
```

Acceptance: Migrations apply cleanly; User, Doctor, Location all expose is_verified; FacilityMembership present.

Phase 2 — Onboarding serializers & atomic registration endpoints

Goal: Self-serve registration for Facility Admins and Doctors that creates all related records atomically and leaves them unverified.

Files

• **NEW** accounts/serializers_onboarding.py • accounts/views.py • accounts/urls.py

Serializers & Endpoints

FacilityRegistrationSerializer — facility_type, name, license_number, cascading location, phone, password. Atomic create of User (facility_admin) + Location + detail + FacilityMembership. All is_verified=False.

DoctorRegistrationSerializer — name, bmdc_number, specialty, qualification, phone, password. Atomic create of User (doctor) + Doctor. All is_verified=False.

Endpoints: POST /api/v1/auth/register/facility/ and /register/doctor/ (AllowAny + throttle).

Verification

```
cd doctors_hub_backend && python manage.py check
```

Acceptance: Both endpoints create the full object graph in one transaction and leave everything unverified.

Phase 3 — Facility Staff delegation endpoints

Goal: Facility Admins (and Super Admins) can add/remove staff for their own locations only.

- GET/POST /api/v1/facilities/{location_id}/staff/
- DELETE /api/v1/facilities/{location_id}/staff/{user_id}/

Permission: Super Admin **or** Facility Admin with membership role=admin on that location. Cross-facility → 403.

Verification

```
pytest tests/test_staff_delegation.py -q
```

Acceptance: Facility Admin manages only own staff; foreign location → 403; Super Admin can manage any.

Phase 4 — Super Admin Verification Queue & Platform Admin management

Goal: Super Admin 1-click approve/reject + ability to invite other Super Admins.

- GET /api/v1/admin/verifications/
- POST /api/v1/admin/verifications/{id}/approve/ and /reject/
- GET/POST /api/v1/admin/platform-admins/ (Super Admin only — never public)

Approve sets is_verified=True. Reject may store reason and soft-deactivate.

Verification

```
pytest tests/test_verification.py -q
```

Acceptance: Only Super Admins can list/approve; after approval the entity becomes publicly visible.

Phase 5 — Automated tests

Create tests/test_onboarding.py, test_staff_delegation.py, test_verification.py covering:

- Facility registration creates full graph with is_verified=False
- Doctor registration creates User + Doctor with BMDC, is_verified=False
- Duplicate phone / BMDC → 400
- Facility Admin can add staff to own location → 201; foreign → 403
- Super Admin 1-click approve flips is_verified=True
- Non-Super-Admin cannot hit verification endpoints → 403
- Super Admin can create another Super Admin

Verification

```
pytest tests/test_onboarding.py tests/test_staff_delegation.py tests/test_verification.py -q
```

Phase 6 — Frontend: Unified Admin Gateway (Registration Hub)

Transform AdminLoginForm.jsx into multi-tab hub: Sign In | Register Diagnostic Center | Register Hospital | Register Doctor. Cascading location selectors, BMDC + specialty selection. Success message: can log in immediately, appears in public search only after verification.

Acceptance: Logged-out /admin shows four tabs; registration allows login; entity hidden until verified.

Phase 7 — Frontend: StaffTab + VerificationQueueTab + Nav updates

- **NEW** StaffTab.jsx — Facility Admin + Super Admin; table + Add Staff modal + revoke
- **NEW** VerificationQueueTab.jsx — Super Admin only; pending tables + 1-click Verify/Reject + badge
- Update AdminNavStrip.jsx and AdminContext for role-based visibility

Acceptance: Correct role sees correct tabs; non-privileged roles never see Staff or Queue tabs.

Phase 8 — Hardening, throttling, audit & cleanup

1. DRF throttling on public registration endpoints
2. Public catalog querysets filter is_verified=True
3. Basic audit logging for verification actions
4. Remove leftover AllowAny on sensitive admin endpoints
5. Update README / AGENTS.md
6. Full test suite + manual end-to-end walkthrough

Verification

```
cd doctors_hub_backend && pytest -q
```

Recommended Antigravity CLI Execution Order

```
# Inside the repo root
agy
```

```
# Feed phases one at a time:
Implement Phase 1 of ONBOARDING_IMPLEMENTATION_PLAN.md exactly.
Stop after verification passes.
```

```
# After success:
Implement Phase 2 of ONBOARDING_IMPLEMENTATION_PLAN.md exactly. ...
```

Use /rewind or /undo freely if a phase introduces regressions. Prefer small, reviewable diffs; approve file changes via the artifact review screen (Ctrl+R) before continuing.

Out of Scope (future work)

- Email / SMS notifications on approval or rejection
- Real BMDC / DGHS external API verification
- Multi-branch staff assignment UI polish
- Soft-delete + archive instead of hard deactivation
- Advanced risk scoring on the verification queue

End of plan. When all phases pass verification, the platform supports fully decentralized onboarding for 500+ doctors and 100+ facilities while keeping Super Admin in a pure governance role.